FINITENESS AND TORSION FREENESS OF THE LIMIT OF THE GLOBAL SECTIONS OF AN INVERSE SYSTEM OF SHEAVES OVER A COMPLETE PRINCIPAL IDEAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Finiteness criterion over a complete Noetherian ring. Stacks Project, Algebra, Lemma algebra-lemma-finite-over-complete-ring (tag 031D). Lifts finiteness of M/fM to finiteness of M using the vanishing of the intersection of the f^nM.
- The topology of the inverse system modulo powers of f. Stacks Project, Cohomology of Sheaves, Lemma lemma-topology-I-adic-f (tag 0EHA). Gives the containment M/fM inside H^0(X, F_1) that starts the proof.
- The f-divisibility condition on the inverse system. Stacks Project, Cohomology of Sheaves, Lemma lemma-equivalent-f-good (tag 0H39), condition (1). Identifies the kernel of F_{n+1} -> F_n, which is what makes f a nonzerodivisor on the limit.

Source reference. Original lemma 0H39: sources/stacks-chow/tags.txt, source lines 21309–21309. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0EHA: sources/stacks-chow/tags.txt, source lines 18125–18125. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 031D: sources/stacks-chow/tags.txt, source lines 3980–3980. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A be a Noetherian ring complete with respect to a principal ideal (f). Let X be a topological space. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

be an inverse system of sheaves of A-modules. Assume

(1) *$\Gamma(X, \mathcal{F}_1)$ is a finite A-module,*
(2) *X, f, $(\mathcal{F}_n)$ satisfy condition (1) of Lemma 0H39.*

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Then

$$M = \lim \Gamma(X, \mathcal{F}_n)$$

is a finite A-module, f is a nonzerodivisor on M, and M/fM is the image of M in $\Gamma(X, \mathcal{F}_1)$.

Proof. By Lemma 0EHA we have $M/fM \subset H^0(X, \mathcal{F}_1)$. From (1) and the Noetherian property of A we get that M/fM is a finite A-module. Observe that $\bigcap f^n M = 0$ as $f^n M$ maps to zero in $H^0(X, \mathcal{F}_n)$. By Algebra, Lemma 031D we conclude that M is finite over A. Finally, suppose $s = (s_n) \in M = \lim H^0(X, \mathcal{F}_n)$ satisfies $fs = 0$. Then s_{n+1} is in the kernel of $\mathcal{F}_{n+1} \to \mathcal{F}_n$ by condition (1) of Lemma 0H39. Hence $s_n = 0$. Since n was arbitrary, we see $s = 0$. Thus f is a nonzerodivisor on M. $\square$